CSE 201 Final

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I. Question (41 points)

Declare three classes named Player, ComputerPlayer, and HumanPlayer to represent players in a Nim game. In the Nim game, there are N coins on the deck in the beginning of the game. Each player (in turn) takes 1 or 2 coins from deck. Whoever takes the last coin wins. The abstract class Player contains:

- (1 pt) A string data field named **name** that specify the name of the player.
- (1 pt) An integer data field named **point** that specify the total point the player has gained.
- (3 pts) One argument constructor **Player(String name)** that sets point to 0.
- (4 pts) Two getter methods for the two fields **name** and **point**.
- (4 pts) Two setter methods for the two fields **name** and **point**.
- (2 pts) Two abstract methods **int play(int N)** and **void display()**.

The class HumanPlayer extends Player by implementing

- (2 pts) One argument constructor that calls its super class's constructor.
- (3 pts) **void display()** that displays the name of the player and its point in the screen.
- (8 pts) **int play(int N)** that takes the number of coins played by the human player as input and returns.

The class ComputerPlayer extends Player by implementing

- (2 pts) One argument constructor that calls its super class's constructor.
- (3 pts) **void display()** that displays "Computer" and its point in the screen.
- (8 pts) **int play(int N)** where the computer takes (3 - N % 3) coins if N % 3 == 1 or N % 3 == 0, takes a random number of coins (1 or 2) if N % 3 == 0. Here N represents the number of coins that is currently on the deck.

II. Question (15 points)

Using the methods and classes declared in the first question, implement the main method which

- (2 pts) Declares two players Olcay and the computer.
- (3 pts) Takes as input the number of coins on the deck.
- (10 pts) Calls play method of the two players until there are no coins left on the deck. Your code will also display the current situation of the game.

III. Question (28 points)

Declare class Point2D to represent points in two dimensional plane. The class Point2D contains:

- (1 pt) An integer field x to represent the x axis value of the point.
- (1 pt) An integer field y to represent the y axis value of the point.
- (3 pts) Two argument constructor **Point2D(int x, int y)**.
- (4 pts) Two getter methods for the two fields **x** and **y**.
- (4 pts) Two setter methods for the two fields **x** and **y**.
- (5 pts) The methods **double length()** which calculates the length of the vector with the formula $\sqrt{x^2 + y^2}$.
- (10 pts) A static method **void sort(Point2D[] points)**, that sorts an array of points in increasing order of length.

IV. Question (16 points)

Declare class Point3D that extends Point2D to represent points in three dimensional plane. The class Point3D contains:

- (1 pt) An integer field z to represent the z axis value of the point.
- (3 pts) Three argument constructor **Point3D(int x, int y, int z)** that calls its parent class constructor.
- (2 pts) Getter method for the field **z**.
- (2 pts) Setter method for the field **z**.
- (8 pts) A static method **void totalLength(Point3D[] points)**, that calculates the sum of the lengths of the points. The length of the point in 3D is calculated as $\sqrt{x^2 + y^2 + z^2}$.